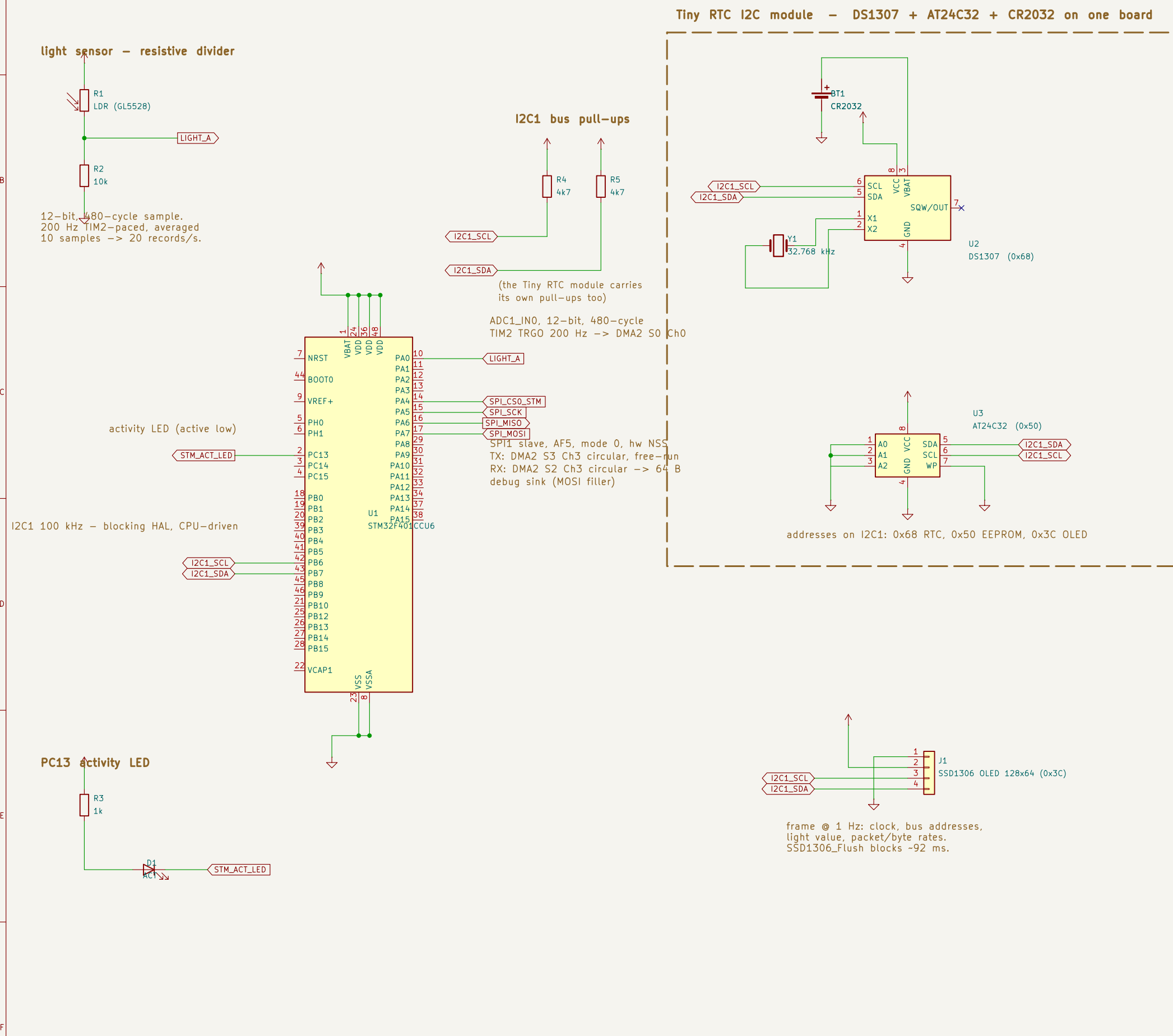


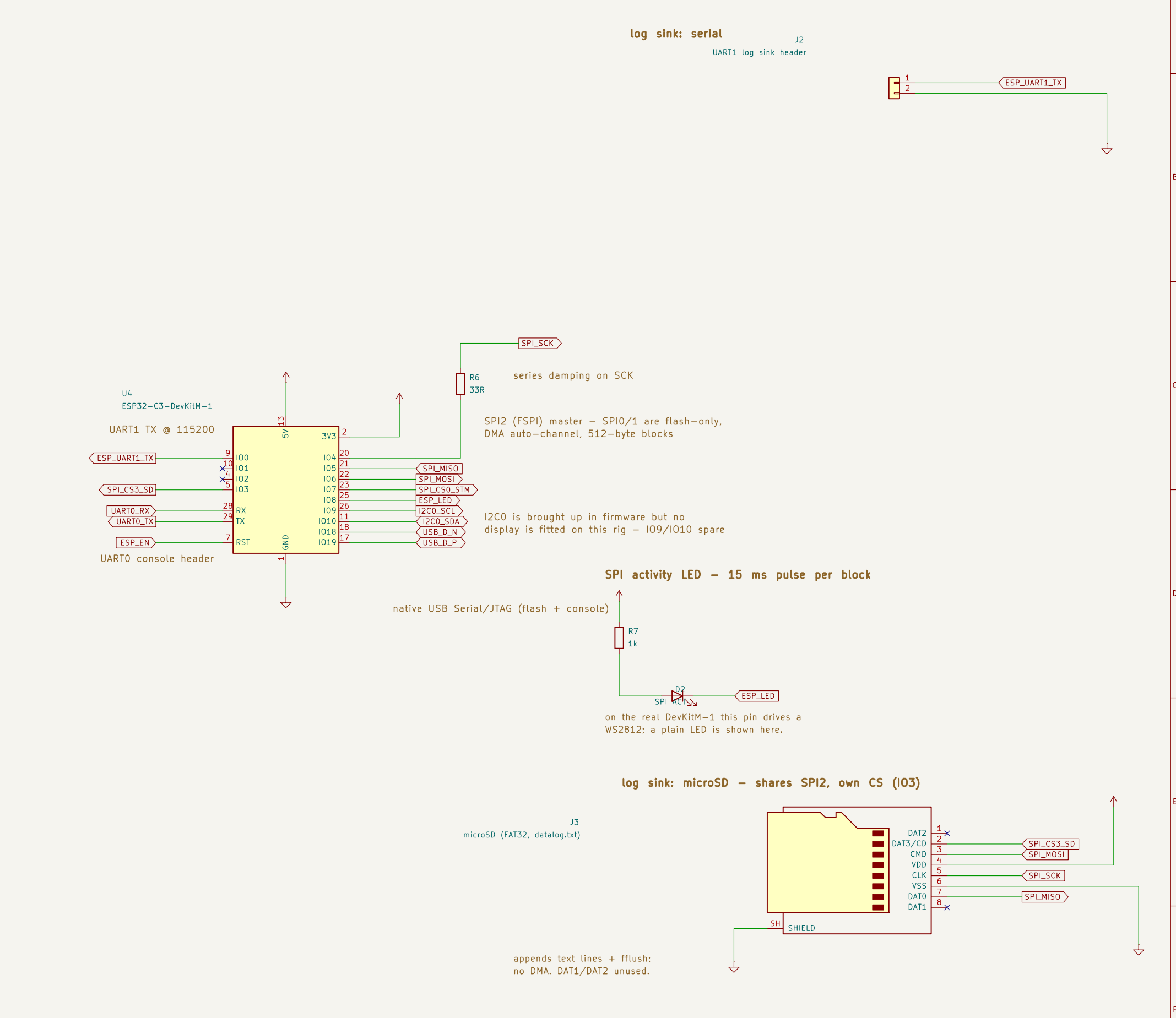
Workshop 4 mini-project – system interconnect

ESP32-C3 SPI master / data-log collector + STM32F401 I2C sensor node acting as SPI log slave. Sources: [espc3/](#) and [stm/miniproject-4/](#)
VISUALIZATION ONLY – block-level interconnect, protocol and DMA annotation. No decoupling, power tree, protection or footprints; not intended for PCB fabrication.

STM32F401CCU6 NODE - I2C application + SPI log slave



ESP32-C3 NODE – SPI master, collector and log sinks



SPI LOG STREAM - MASTER / SLAVE CONTRACT
source of truth: espc3/src/LogProtocol.hpp + LogPacket.hpp
(the STM slave mirrors these constants in Src/log_emission.c)

Topology Star. The ESP32-C3 is the only master; every other node is a slave on the shared SCK/MOSI/MISO with its own CS. Slaves never talk to each other.

The ceiling is the F401 slave, not the master: it runs on HSI with no PLL, so PCLK2 = 16 MHz and SCK wants to stay well under PCLK/4 = 4 MHz.

Packet [18-byte header][payload 0..255 B][CRC-16/CCITT-FALSE]

framing The parser resyncs on the next 0xA5 whose CRC checks out so torn frames at the ring seam and 0xFF idle lines are

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Status 0x00 ENTRY, 0x01 NOENTRY (heartbeat);
      flags 0x80 OVERFLOW (slave dropped), 0x40 NEARFULL.
Records TIME (VT_DATETIME, 1 Hz from the DS1307) and
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Rates The slave emits ~470 B/s; the master drains ~125 KB/s in 512-byte blocks, so a faster clock re-reads the same ring more often – it buys latency margin, not throughput.

STM32F401CCU6 – PERIPHERAL AND DMA MAP
stm/miniproject-4/ (STM32CubeIDE, HAL; comments in Ukrainian)

Clock HSI 16 MHz, no PLL. PCLK2 = 16 MHz.
PA0 ADC1_IN0, 12-bit, 480-cycle sample time.
TIM2 PSC 1599 / ARR 49 -> TRGO on update = 200 Hz.

DMA2 S0 Ch0 ADC1 -> 20-sample circular ring, periph->mem;
half-word, half + full transfer IRQ. No CPU per sample;
each half-buffer is averaged into one LGHT record, so

SPI1 slave PA4 NSS (SPI_NSS_HARD_INPUT), PA5 SCK, PA6 MISO, PA7 MOSI, AF5, mode 0, MSB first, CRC off.

DMA2 S3 Ch3 mem->periph, CIRCULAR, free-running: the TX ring wired straight into SPI1->DR and never stops. The master reads whatever currently sits under the write pointer;

DMA2 S2 Ch3 periph->mem, CIRCULAR: SPI1->DR into a 64-byte
captures the master's MOSI filler. Debug drain only.

I2C1 PB6 SCL / PB7 SDA, 100 kHz, blocking HAL, CPU-drive

PC13 Activity LED, active low, pulsed on SPI traffic.

EEPROM One light byte per hour into the AT24C32; the next-w pointer is stored in the EEPROM itself.

ESP32-C3-DevKitM-1 - PERIPHERAL AND DMA MAP
espc3/ (PlatformIO + ESP-IDF, C++, FreeRTOS)

SPI2 / FSPI SCLK IO4, MISO IO5, MOSI IO6. SPI0/1 are reserved for the flash. DMA on an auto-assigned channel, 512-byte blocks. The sweep is pipelined: device i's block clocks on the wire while device i-1's block is parsed, so at most two devices are attached and only one transfer is

CS scan I07 = the STM32 log slave. drainPresent() runs flat-out (its SPI transfers block and thereby yield the CPU); probeAbsent() runs once every 1000 ms purely to notice

103 hot-plug. Spare CS pins are reserved for a second slave (an ESP32-S3 laser node) – not wired on this rig.
microSD CS. Same shared bus, dedicated CS, SDSPI + FATFS, synchronous – no DMA on this path.

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I2C0    SDA I010 / SCL I009, 400 kHz. Brought up in firmware
but no display is fitted on this rig - the pins are
        spare and no pull-ups are needed.
UART1   TX on I00 @ 115200 -> serial log sink. Kept off UART0

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UART0 IO20 RX / IO21 TX - console header.
USB IO18 D- / IO19 D+ - native USB Serial/JTAG, used for flashing and the console. It re-enumerates on every

Activity LED, active low, 15 ms pulse per block.
Every record fans out to all LogTarget sinks. In use here: UART1 and the microSD card. The OLED sink is still

LogRecord::data points into a shared buffer valid only during write() – a sink that keeps the bytes must copy them first.

Pin limits IO11-17 = SPI flash, IO18/19 = USB, IO20/21 = UART0
IO2/8/9 = strapping. Validate before moving pins.

INTER-BOARD LINK – common ground required

net	ESP32-C3	STM32F401	microSD
SPI_SCK	I04 out	PA5 SCK	CLK
SPI_MOSI	I06 out	PA7 MOSI	CMD (DI)
SPI_MISO	I05 in	PA6 MISO	DAT0 (DO)
SPI_CS0_STM	I07 out	PA4 NSS	-
SPI_CS3_SD	I03 out	-	DAT3/CS

KNOWN ISSUES AND DESIGN NOTES (from README.md)

A failed wire once produced wrong data over SPI, kept the address stuck in 0x5830 and the data stuck in 0x0000 for 2 or 3 clock margin. Step back to the MHz 250–500 kHz if you suspect the symptom is CRC-rejected frames, which look like a silent fall in recorded rate rather than an error.

When the SPI master is not busy, whenever the STM32 was in power, SPI1 uses hardware NSS, and until the collector's first attempt that CS pin was left floating, so noise on it could make the master misinterpret the data and corrupt it. Fixed in hardware by driving every CS high right after bus init; a hardware pull-up on each CS/NSS line is still required, since the CS pin is not pulled up by default.

IC21 needs external pull-ups (R4/R5); the Tiny RTC module already carries its own. IC20 has no device fitted.

The two boards are not connected to the power supply, tie the grounds together or the link will not work.

SCOPE: visualization only. No decoupling, power tree, ESD or level shifting, and no footprints are shown. Do not fabricate from this sheet.

Visualization only – not for PCB fabrication

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Title: Workshop 4 mini-project – ESP32-C3 log collector + STM

Size: A2	Date: 2026-08-01
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